

Hello and thank you for participating in our work!

You are an important part of a project in the Stenglein Lab to assess the prevalence of a virus that infects the common fruit fly, *Drosophila melanogaster*. The data from these collections will help to uncover the mechanisms behind virus-host interactions that drive evolution. We hope that you will want to continue to be part of this project over several years to extend our knowledge of the virus over time and space and we welcome your feedback about how we can improve these instructions or the project overall.

We also hope you enjoy this process and have fun with it! If you have any questions, please contact Lexi through the forum set up (*Fly Trap Q&A*) on the CitSci.org platform.

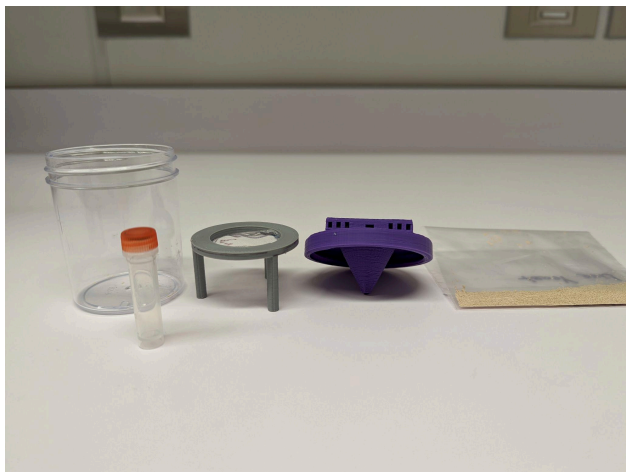
What you'll find in this document:

1. Step by step instructions on setting up your fly trap and collecting flies.
2. Instructions on how to prepare the samples for shipping back to our lab.

*Please find the instructions to the fly eyesight experiment on the resources page found on the CitSci.org webpage (<https://citsci.org/projects/virus-tracking-in-fruit-flies>)

**Each kit comes with three return envelopes and 2 mL tubes for the collection of specimens over several points during the collection period. Please use all of these, waiting 1-2 weeks between collections.

Package Contents :



- 1 Jar
- 1 3D printed “Flyght House” fly trap lid
- 1 3D printed mesh table
- 1 Bag of yeast
- 3 2 mL tubes
- 3 Pre-labeled return envelopes

* Instructions for the optional experiment and the materials required can be found in the resources page on the CitSci.org website:

<https://citsci.org/projects/virus-tracking-in-fruit-flies>

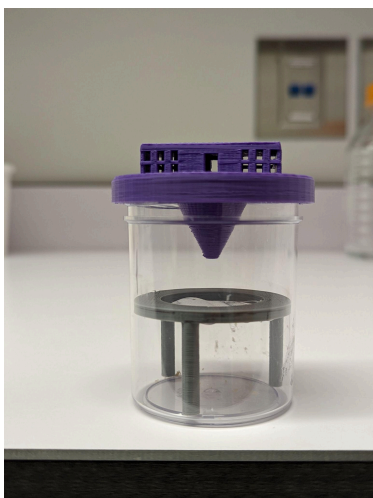
What you provide

- Banana

Instructions for Setting Up the Fly Trap

1. Add a small slice of banana (the thickness of a few quarters stacked on top of each other) to the bottom of the empty jar and sprinkle a very small pinch of yeast on top of the banana.
2. Place the “table” into the jar (it will be snug). If the banana touches the mesh on the “table” you have put too much banana in, please take some out before continuing.
3. Screw on the fly trap lid (flyght house) to the jar

Example:



4. Place the jar in a location where you expect to find fruit flies. Some good locations include in the kitchen next to fruit or outside next to a compost pile, fruit tree, or garbage can.

*If placing the jar outside try to put it somewhere that it won't be disturbed by other people or animals.

5. Keep an eye on the trap for up to 2 weeks but if your trap collects 12 or more flies before then you may stop collecting early. Below is an image of the type of fruit fly we are looking for. Other insects may get in the traps. That is okay, we will identify everything that gets to us.



https://commons.wikimedia.org/wiki/File:Drosophila_melanogaster_Proboscis.jpg

6. Follow the instructions in the next section to return flies to us.

** If you're not sure if the trap is ready, please contact Lexi through the CitSci.org platform for assistance!*

Please view the following video for instructions in an alternative format:

https://www.youtube.com/watch?v=ZyKi_uuAkJA

Instructions for shipping flies back to our lab:

1. Once you are ready to end the experiment you will need to place them in one of the small sample tubes from your kit.
2. Take the jar and place it in your freezer for **1 hour** (If you do less than 1 hour, the flies may wake up while you try to move them!).
3. Remove the jar from the freezer and gently tap it on a counter so the flies fall to the bottom of the trap (i.e. on top of the mesh table).
4. Remove the lid and gently tap the flies onto a clean paper towel and allow them to dry for **1 day**. Allowing the flies to dry out before shipping will keep them from sticking together in the tube.
5. Once the flies are dried, gently transfer them into one of the small sample tubes provided in your kit. Please label the tube with your name, the date the flies were collected, and location.
6. Place the tube into the prepaid and addressed envelope you received in your kit and mail it like you would a normal letter!

***Remember to fill out the metadata sheet on the CitSci.org website with the general location of collection, date of collection and number of flies!**

Please view the following video for instructions in an alternative format:

https://www.youtube.com/watch?v=ZyKi_uuAkJA

Thank you so much for participating! We can't wait to see the flies and hear the feedback from you! Additionally, we plan to update the CitSci.org platform with our results as samples begin to be processed, be sure to look back on occasion to see what viruses we found in your flies!